

**Equipment Procedure
for Determination Butylated Hydroxy Toluene in Transformer Oil by Gas
Chromatography-FID**

Equipment Number: EQ-66

Serial Number: 722121237



1. PURPOSE

This standard test method procedure is intended to provide chemist / analyst / technician the general guide on performing the test analysis appropriately.

2. SCOPE

This test method covers the determination of special additives which is BHT or called DBPC = 2,6-di-tert-butyl-para-cresol in used and unused insulating liquids.

3. PRINCIPLE OF ANALYSIS

1 ml of oil sample is diluted with 2 ml of IPA or Toluene solvent which spiked already with internal standard (2, 4, 6-Trimethylphenol). A microliter of oil sample is injected in the GC-FID. The peak area of the BHT in the chromatogram is integrated and compared with the related area in the standard calibration curve to calculate the concentration in part per million (ppm).

4. SAFETY PRECAUTIONS

5.1 Use a proper electrical outlet of correct voltage and current handling capacity.

5.2 Disconnect from the power supply prior to maintenance and servicing.

5.3 Ensure that the operating switch operates properly.

5.4 Lab coat, Goggles, and gloves are a must.

5.5 Working under fume hood is a must.

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5. DISTRIBUTION

This operational procedure is distributed to the following:

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General Manager

Laboratory-In-Charge/Technical Manager, Chemistry Laboratory

6. TERMS AND DEFINITIONS

Butylated hydroxytoluene (BHT), also known as dibutylhydroxytoluene, is a lipophilic organic compound, chemically a derivative of phenol, that is useful for its antioxidant properties having molar mass 220.36 g/mol and boiling point 265° C.

Flame Ionization detector (FID): A Flame ionization detector (FID) consists of a hydrogen (H₂)/air flame and a collector plate. The effluent from the GC column passes through the flame, which breaks down organic molecules and produces ions. The ions are collected on a biased electrode and produce an electrical signal.

7. APPARATUS / REAGENTS

6.1 Gas Chromatography (GC-FID)

6.2 Analytical Balance

6.3 Helium, Hydrogen and Air Cylinder

6.4 BHT 98.0%

6.5 2,4,6 Trimethylphenol (Internal Standard)

6.6 Iso Propyl Alcohol analytical grade

6.7 2ml Vial with TFE-fluorocarbon Faced Butyl Septa cap

6.8 5ml Micropipettes with pipette tip

8. PROCEDURE

8.1. Preparation of Standards / Internal Standard

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8.1.1. Preparation of Stock BHT Standard Solution (10000 ppm w/w).

8.1.2. Weigh 0.102 (± 0.0010 mg) of BHT in 200ml beaker and add few drops of IPA just to dissolve it and dilute with BHT free oil up to 100 g.

8.1.3. Prepare the following standard concentration of BHT (1000ppm, 2000ppm and 4000ppm by weight) from the BHT stock solution 10000 ppm standard and dilute to the final weight with BHT free oil.

	Concentration	Weight (g) (Stock Sol'n.)	Final Weight with oil (g)
1	1000 ppm	10 (± 0.010 mg)	100
2	2000 ppm	20 (± 0.010 mg)	100
3	4000 ppm	40 (± 0.010 mg)	100

8.1.4. Preparation of Internal Standard Solution (In IPA).

Weigh 0.40 (± 0.10 mg) of 2,4,6-trimethylphenol in 100ml flask and dissolve it with IPA up to 50 gm of IPA solution.

8.2. Preparation of Std Calibration

7.2.1 Take 1 ml of each standard and add 2 ml of internal standard using calibrated pipette into 15mL tube with cap.

7.2.2 Transfer to 2 mL vial close the vial and shake.

7.2.3 Close the vial and shake.

7.2.4 Inject the standards in GC-FID on already set up method "BHT IEC 60666".

7.2.5 Change type to "Calibration Standard" and level for each standard.

7.2.6 Evaluate the chromatograph and calibrate the method.

8.3. Preparation of Sample

7.2.1 Take 1 ml of each standard and add 2 ml of internal standard using calibrated pipette into 15mL tube with cap.

7.2.2 Transfer to 2 mL vial close the vial and shake.

7.3.2 Set up method "BHT IEC 60666" on GC system.

7.3.3 Prepare a sequence, 1st vial Solvent Blank, 2nd Vial QC BHT.

7.3.4 Check QC in the QC chart then add all other samples to be analyzed in sequence.

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7.3.5 After analysis is completed, integrate the peaks and record the results up to three decimals.

7.3.6 Run "Idle" method at the end of the sequence.

9. Calibration, Verification and Cleanup

- 9.1. A yearly commissioning service preventive maintenance is provided by Naizak Engineer for highly technical checkup.
- 9.2. Column Check Up
 - 9.2.1. Condition the column monthly using "Column condition" method.
 - 9.2.2. Run wash between the sample sequence to prevent any sample accumulation.
- 9.3. Verification is done by using standard prepared in-house or a sample of known concentration and recording in the quality control chart.
- 9.4. Column conditioning is performed every week.

10. REFERENCES

- 10.1. IEC 60666-2010 Detection and determination of specified additives in mineral insulating oils.
- 10.2. Thermo Scientific Trace 1610 manual

11.

Method	FID temperature	Oven Program	Carrier Gas	Injected volume
BHT_IEC60666	300C	Initial Temp: 180C hold for 1min Ramp 15C/min to 245C then hold for 0min Ramp 50C/min to 300C then hold for 16.6min	Helium 2ml/min	1uL

12. ATTACHMENTS

- 12.1. EQ "Equipment Operation Record"
- 12.2. QF "Equipment Maintenance Record"

13. RECORDS

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10.1 EQ-302 GC Clarus-Operation Records

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